

Euclid's Elements — Book I

Proposition 23

Formal Reconstruction — Technical Documentation

Jurek Rózański
Military University of Technology, Warsaw, Poland
`jerzy.rozanski@wat.edu.pl`

CGJTEAMLAB

Copying a given angle at a given point

1 Introduction

Euclid's Proposition I.23 asks for the construction, at a prescribed point on a prescribed straight line, of an angle congruent to a given rectilinear angle.

In Euclid's development this is a genuine construction theorem. The angle is not transported directly. Euclid chooses a triangle containing the given angle, reconstructs a second triangle from its three side lengths using Proposition I.22, and then invokes Proposition I.8 (SSS) to identify the required angle.

In the Hilbert reconstruction used by CGJteamLab the logical status is different. Hilbert's congruence axiom III.4 already provides angle construction on a prescribed side of a prescribed ray, together with uniqueness of the resulting ray. Consequently the formal Proposition I.23 is a thin wrapper over a stronger foundational principle.

This contrast is mathematically important. At I.23 the textual order of Euclid's propositions and the dependency order induced by the Hilbert foundation visibly diverge.

2 The Classical Configuration

Euclid's construction is best displayed after the auxiliary triangles have actually been constructed. Let the given angle be represented by triangle CDE . On the prescribed straight line through A , choose G , and construct a second triangle AFG so that

$$AG \cong CE, \quad AF \cong CD, \quad FG \cong DE.$$

The point G lies on the prescribed line through A .

The figure records the actual constructional state, not merely two already-congruent angles.

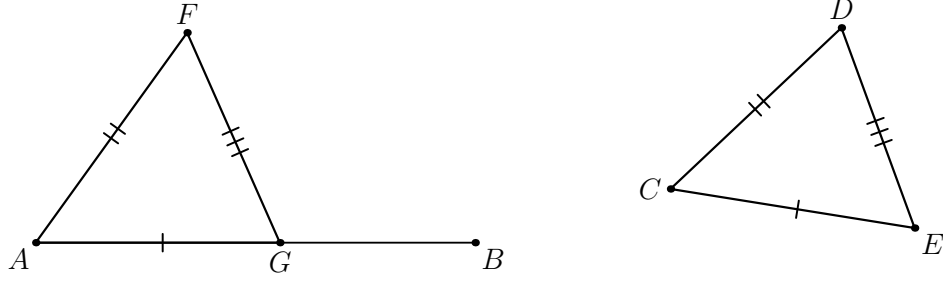


Figure 1: Classical Euclidean auxiliary construction for I.23. Proposition I.22 supplies triangle AFG with the three side congruences corresponding to triangle CDE . Proposition I.8 then identifies the transported angle.

3 Classical Proof

Euclid derives angle transport from earlier propositions rather than taking it as primitive. Choose points on the two arms of the given angle, thereby obtaining a triangle and three side lengths. The triangle inequalities needed for the construction are supplied in the classical dependency chain by Proposition I.20. Proposition I.22 then constructs a triangle with the same three side lengths on the prescribed line. Proposition I.8 gives SSS congruence, hence equality of the corresponding included angles.

In the notation of Figure 1,

$$AG \cong CE, \quad AF \cong CD, \quad FG \cong DE$$

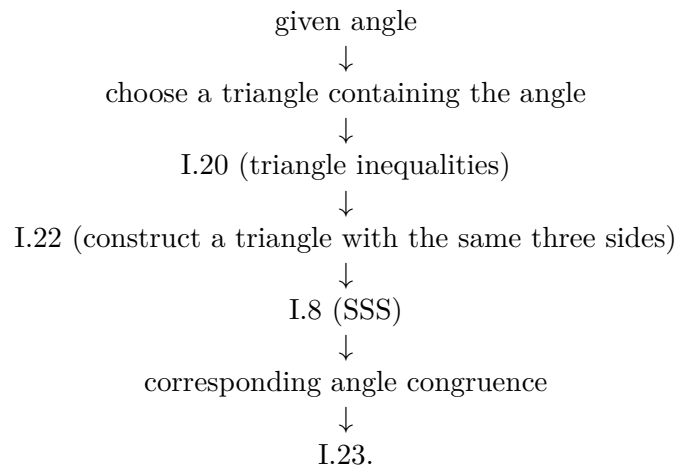
implies

$$\triangle AFG \cong \triangle CDE,$$

and therefore

$$\angle FAG \cong \angle DCE.$$

The classical dependency pattern is therefore



Thus Euclid reduces angle transport to segment construction, circle intersection, and triangle congruence.

4 Hilbert III.4

Hilbert reverses the foundational emphasis. Instead of deriving angle transport from triangle construction, the congruence axioms contain a primitive angle-construction principle. In the present formal layer the relevant operation is

`HilbertCongruence.angle_construction`

Conceptually, its input consists of

- a nondegenerate source angle;
- a prescribed initial ray;
- the supporting line of that ray;
- a choice of one side of the supporting line.

It returns a ray on the selected side forming an angle congruent to the source angle. The Hilbert operation is stronger than Euclid I.23 because it also contains a uniqueness assertion for that ray.

5 Proof Architecture of the Reconstruction

5.1 The Source Angle

The hypothesis

$$\neg \text{PrimCollinear}(A, B, C)$$

ensures that $\angle ABC$ is nondegenerate.

5.2 The Prescribed Ray

The points D and E represent the initial ray of the new angle, with vertex E and initial ray ED . The formal hypotheses say that

$$D \neq E$$

and that both points lie on a common supporting line **base**. Thus the formal input specifies an oriented ray rather than only an unoriented line.

5.3 Selecting the Side

A point S not lying on **base** selects one of the two sides of that line. The constructed point F must satisfy

$$\text{HilbertSameSide}(F, S; \text{base}).$$

This datum has no explicit counterpart in Euclid's informal formulation: the intended side is normally supplied by the diagram. In an axiomatic reconstruction it must be stated.

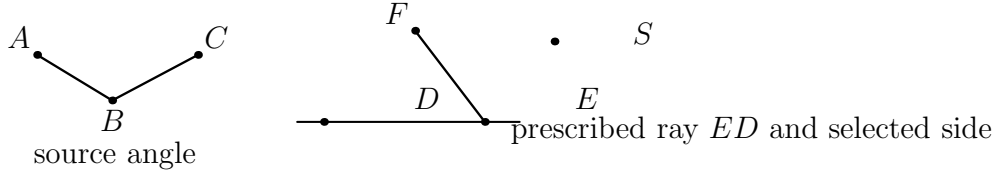


Figure 2: Hilbert III.4 configuration. The source angle is $\angle ABC$; ED is the prescribed initial ray on the supporting line; S selects a side of that line; and the constructed point F lies on the selected side with $\angle ABC \cong \angle DEF$.

5.4 Applying Angle Construction

Hilbert III.4 supplies the new point F directly. The formal geometric state is shown in Figure 2.

The uniqueness component supplied by III.4 is not needed for the Euclidean proposition and is discarded by the public wrapper.

6 Lean Formalization

The current production theorem is

```

theorem euclid_proposition_23
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D E S : Geo.Point)
  (hABC : Not (PrimCollinear Geo A B C))
  (hDE : D != E)
  (base : Geo.Line)
  (hDbase : HilbertIncidence.OnLine D base)
  (hEbase : HilbertIncidence.OnLine E base)
  (hSbase : Not (HilbertIncidence.OnLine S base)) :
  Exists fun F : Geo.Point =>
    HilbertSameSide Geo F S base /\
    Geo.AngleCongruent A B C D E F := by
rcases
  HilbertCongruence.angle_construction
    (Geo := Geo)
    A B C D E S
    hABC hDE base hDbase hEbase hSbase with
  <F, hFSame, hAngle, _hUnique>
exact <F, hFSame, hAngle>

```

The proof is therefore a direct interface extraction:

$$\begin{array}{c}
 \text{source angle} + \text{prescribed ray} + \text{side selector} \\
 \downarrow \\
 \text{HilbertCongruence.angle_construction} \\
 \downarrow \\
 F, \text{ SameSide}(F, S; \text{base}), \quad \angle ABC \cong \angle DEF \\
 \downarrow \\
 \text{euclid_proposition_23.}
 \end{array}$$

There are no proposition-local helper theorems, no geometric case split, and no auxiliary

construction beyond the returned point F .

7 The Euclidean Route and the Hilbert Route

7.1 The Euclidean Route

Euclid reduces angle transport to triangle construction:

$$\text{I.22} + \text{I.8} \longrightarrow \text{I.23},$$

with Proposition I.20 entering the expanded classical route to establish the triangle inequalities required by I.22.

7.2 The Hilbert Route

In the Hilbert foundation the dependency is simply

$$\text{III.4} \longrightarrow \text{I.23}.$$

Angle transport has already been placed in the congruence layer.

7.3 Foundational Interpretation

This is not merely a shorter proof of the same dependency graph. Euclid treats the ability to copy an angle as something constructed from more elementary segment operations. Hilbert treats it as part of the primitive congruence structure. The same visible Book I proposition therefore occupies two different logical levels in the two foundations.

8 Relationship with Earlier Propositions

8.1 Proposition I.8

I.8 is the final congruence theorem in Euclid's proof: once the auxiliary triangle has the required three side lengths, SSS identifies the transported angle. The current Hilbert proof does not call I.8.

8.2 Proposition I.22

Euclid uses I.22 to construct the auxiliary triangle with prescribed sides. The current Hilbert proof does not call the theorem `euclid_proposition_22`. The source module does import `Proposition22`, but that module import must not be confused with a direct theorem-level dependency of the proof body. No import-removal test was performed in this audit.

8.3 Book I Order versus Formal Dependency

The file for I.23 naturally follows I.22 in the Book I sequence, but the formal proof graph contains no theorem-level edge

$$\text{I.22} \longrightarrow \text{I.23}.$$

This is one of the clearest early examples where textual order and formal dependency order diverge.

9 Hilbert and Book Zero Components Used

9.1 HilbertCongruence.angle_construction

This is the direct engine of the proof. It supplies existence, the same-side condition, angle congruence, and a uniqueness property stronger than the public conclusion.

9.2 HilbertSameSide

The same-side relation makes explicit which of the two mirror-image constructions is intended. The point S is therefore genuine geometric input, not bookkeeping.

9.3 HilbertSameRay

The uniqueness clause returned by III.4 is naturally expressed through ray equality / same-ray information. The public theorem does not expose this stronger result.

9.4 No New Book Zero Lemma

Proposition I.23 exposes no missing low-level operation. Unlike several preceding propositions, the exact construction already exists in the Hilbert congruence layer. No new proposition-specific Book Zero lemma is introduced by the current proof.

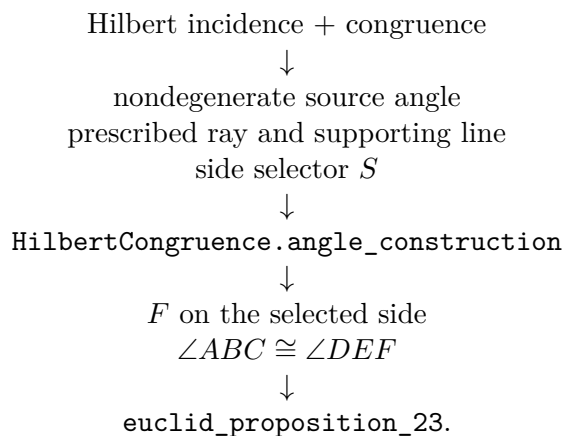
10 Uniqueness and the Strength of the Hilbert Statement

Euclid asks only for a construction. Hilbert III.4 gives existence together with uniqueness on the selected side of the supporting line. If another point F' on that side produces an angle congruent to the same source angle, the corresponding ray must coincide with the ray through F .

The public theorem intentionally forgets this information. This is an interface-design choice: the Book I theorem states the Euclidean conclusion, while the stronger foundational theorem remains available separately.

11 Dependency Summary

The current formal dependency structure is



No direct theorem-level use is made of Proposition I.22 or Proposition I.8, and no new continuity argument is introduced by I.23.

12 Formalization Notes

The point S is additional explicit data relative to Euclid’s informal statement. Without it there are two mirror-image rays satisfying the angle-congruence requirement.

The conditions that D and E lie on **base**, together with $D \neq E$, specify the nondegenerate initial ray ED with vertex E . The noncollinearity of A, B, C guarantees a nondegenerate source angle. The theorem returns a point F , not an abstract ray, matching the point-based ray representation of the Hilbert layer.

The short Lean proof is not evidence that the proposition has become geometrically empty. Its mathematical content has been moved into the axiomatic congruence interface.

13 Diagrammatic Audit

The preserved Book1R chapter contained three figures. The audit retains only geometric states that add new construction data.

- The old Figure 23.1 showed only the initial segment and the given angle. It is removed as a separate float because the completed classical construction already contains these data together with the genuinely new auxiliary triangle.
- The old Figure 23.2 is retained, renumbered as Figure 1. It records the actual Euclidean construction: a new triangle and three side congruences.
- The old Figure 23.3 is retained in role but corrected in orientation, and renumbered as Figure 2. The current Lean target is $\angle DEF$, whose vertex is E ; therefore the prescribed initial ray is ED , not a ray based at D . The corrected figure records the supporting line, side selector S , and the constructed point F .

- No separate uniqueness diagram is added. The uniqueness result changes logical information about the already constructed ray but introduces no new geometric state.

Thus the final audited chapter has two figures: one for Euclid's derived construction and one for Hilbert's primitive angle transport.

14 Conclusion

Proposition I.23 provides a particularly clean example of the difference between historical construction order and formal axiomatic dependency.

Euclid constructs the transported angle indirectly through triangle construction and SSS. The Hilbert reconstruction invokes III.4 directly and then forgets the extra uniqueness information. The audited diagrams now mirror exactly those two geometric routes, without a redundant float for the initial data alone.